

YTU Matematik Mühendisliği Ders Notları

Onur Turan

İçindekiler

Bölüm 1

Advanced Calculus - 1

1.1

1.2 Hafta 1: Vektörler ve Temel Kavramlar

Definiton 1.1. A function of two variables is a rule that assigns a real number W sometimes write: $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$; for instance, $f(x, y) = x^2 + y^2$ or $g(x, y) = x^2 - e^y$.

Likewise, a function of three variables is a rule that assigns a real number to each ordered triple (x, y, z) in its domain $D \subset \mathbb{R}^3$. We sometimes write: $f : D \subset \mathbb{R}^3 \rightarrow \mathbb{R}$; for instance, $f(x, y, z) = x^2 + y^2 + z^2$ or $g(x, y, z) = x^2 - e^{y+z}$.

$f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is a function of n variables if it assigns a real number to each ordered n -tuple $(x_1, x_2, \dots, x_n)$ in its domain $D \subset \mathbb{R}^n$.

Example. (Lesson: Advanced Calculus - 1) (Week: 1) (Topic:)

Find the domain of the functions.

Example 1.1 $f(x, y) = \ln 4 - x^2 - 4y^2$

Example 1.2 $g(x, y) = \sqrt{x^2 + y^2 - 1} + \sqrt{4 - x^2 - y^2}$

Example 1.3 $f(x, y) = \frac{1}{\sqrt{4x^2 - y^2}}$

Definiton 1.2. A neighborhood of a point $\mathbf{a} \in \mathbb{R}^n$ is a set that contains all points $\mathbf{x}$ that are "close" to $\mathbf{a}$. More formally, we can define a neighborhood of $\mathbf{a}$ as follows:

Let $r > 0$. The open ball of radius r centered at $\mathbf{a}$ is defined as:

$$B_r(\mathbf{a}) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{a}\| < r\}$$

A neighborhood of $\mathbf{a}$ is any set that contains an open ball centered at $\mathbf{a}$.

Definiton 1.3. The interior of a set $A \subset \mathbb{R}^n$, denoted $\text{int}(A)$, is the largest open set contained in A . In other words, a point $\mathbf{x} \in A$ is in the interior of A if there exists a neighborhood of $\mathbf{x}$ that is entirely contained in A .

Definiton 1.4. The boundary of a set $A \subset \mathbb{R}^n$, denoted ∂A , is the set of points that can be approached both from inside A and from outside A . Formally, a point $\mathbf{x} \in \mathbb{R}^n$ is in the boundary of A if every neighborhood of $\mathbf{x}$ contains at least one point in A and at least one point not in A .